

THE XOR PROBLEM

What is a Perceptron?

* A Perceptron in ML is the most basic building block of an artificial neural network (ANN).

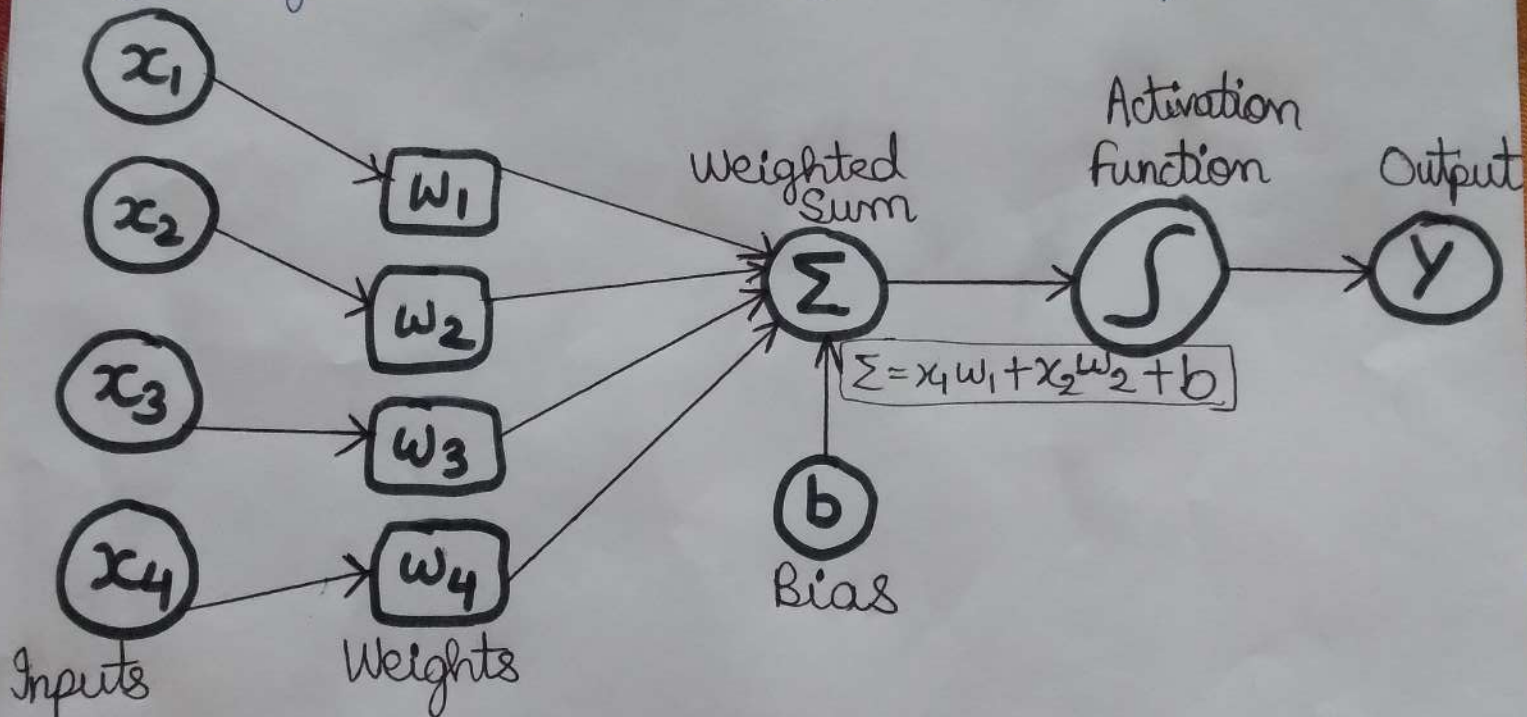
* It processes information through 4 core components:

① Inputs (x)

③ Bias (b)

② Weights (w)

④ Activation function (S)



AND Gate

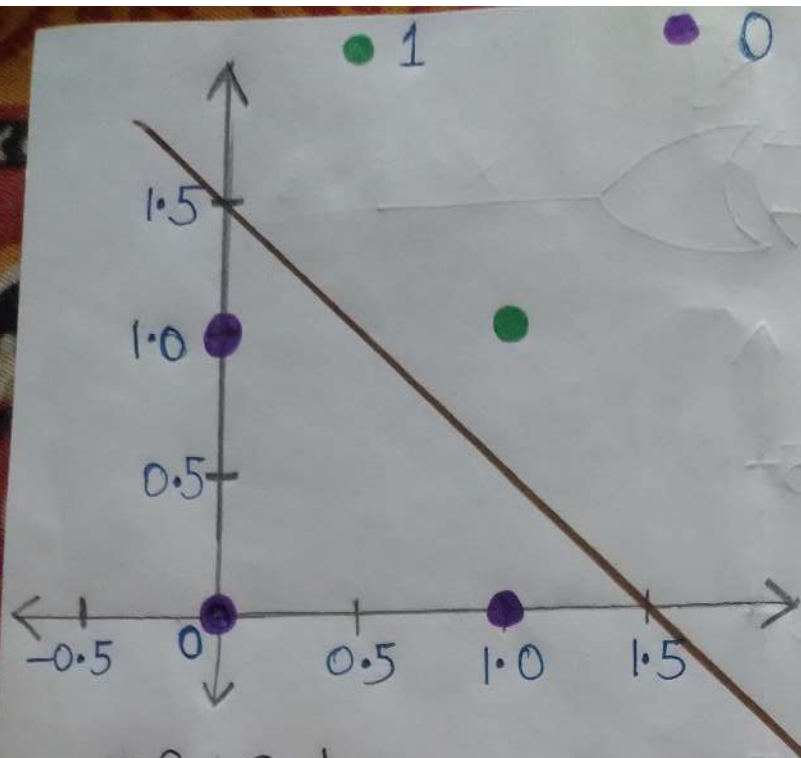
$$X = A \cdot B$$



A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

Outputs 1 only if both inputs are 1.

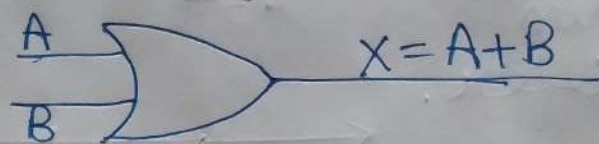
Truth Table



AND Gate is linearly separable means we can easily separate the 0's and 1's using a straight line.

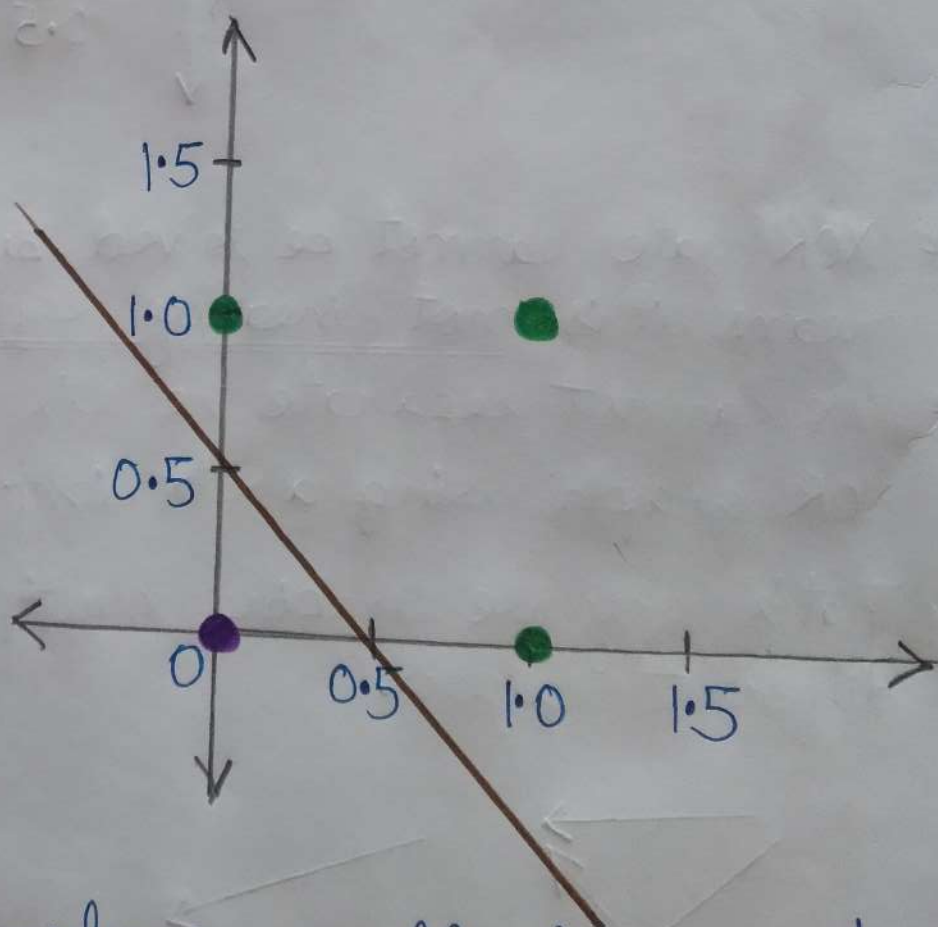
OR Gate

$$X = A + B$$



A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Truth Table



Output 1 if either input is 1.

OR Gate is also linearly separable because it is easy to separate the 0's from 1's using a straight line.

XOR Gate [Exclusive OR]

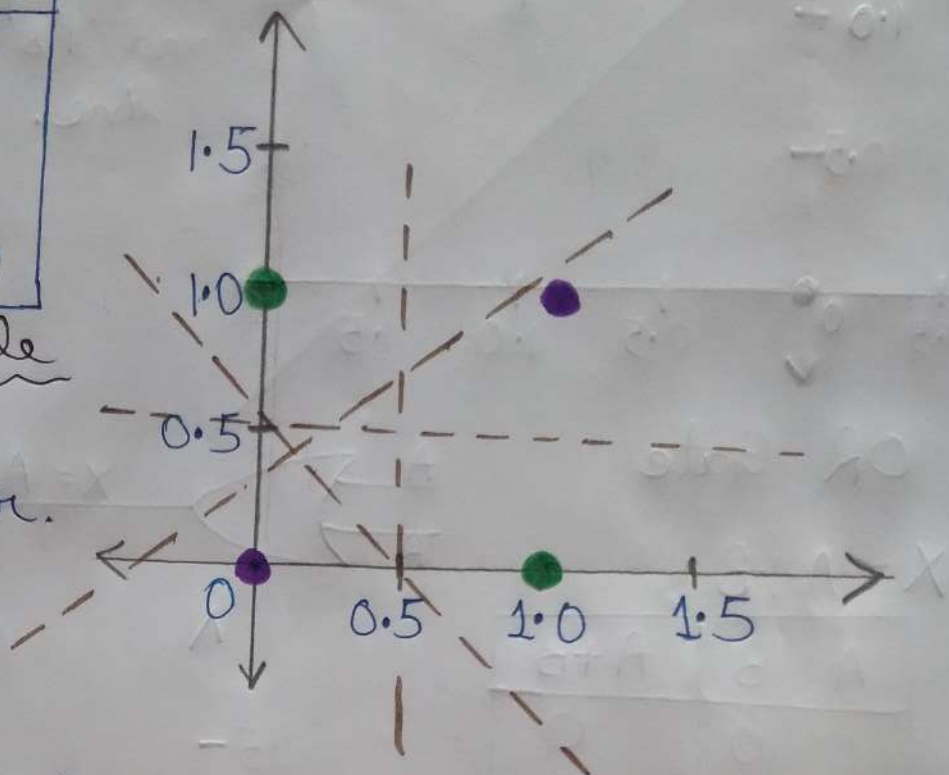
$$X = A \oplus B$$



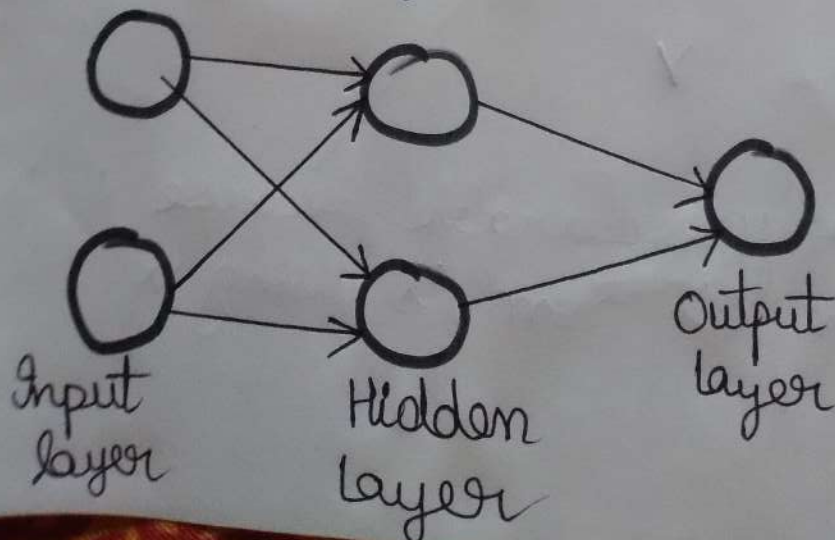
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Truth Table

Output 1 when inputs differ.



- * XOR gate cannot be solved by a single perceptron means it is not linearly separable.
- * We cannot separate the 0's from the 1's in XOR graph using a straight line.
- * That's why we combine multiple perceptrons into different layers.



* Output of 1 perceptron becomes input to another in MLP (Multiple Layer Perceptron).